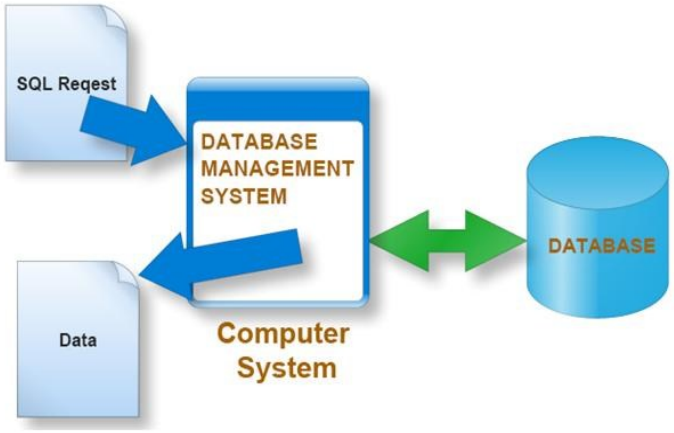


3 Introduction to SQL

Woottikarn Hongwiengchan

SQL

- Structured Query Language
- Standard language for accessing databases.
- ภาษามาตรฐานที่ใช้ในการเข้าถึงฐานข้อมูล เป็นภาษาเชิงโครงสร้าง



SQL statements

Persons

P_Id	LastName	FirstName	Address	City
1	Hansen	Ola	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Kari	Storgt 20	Stavanger

SELECT * FROM Persons

Select * From persons

SQL is not case sensitive
SQL ไม่สนว่าจะตัวใหญ่ หรือตัวเล็ก

Data Manipulation Language (DML)

- **SELECT** - extracts data from a database
- **UPDATE** - updates data in a database
- **DELETE** - deletes data from a database
- **INSERT INTO** - inserts new data into a database

Data Definition Language

- **CREATE DATABASE** - creates a new database
- **ALTER DATABASE** - modifies a database
- **CREATE TABLE** - creates a new table
- **ALTER TABLE** - modifies a table
- **DROP TABLE** - deletes a table
- **CREATE INDEX** - creates an index (search key)
- **DROP INDEX** - deletes an index

SQL - Select

SELECT column_name(s)
FROM table_name

Persons

SELECT * FROM table_name

P_Id	LastName	FirstName	Address	City
1	Hansen	Ola	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Kari	Storgt 20	Stavanger

SELECT LastName,FirstName FROM Persons

LastName	FirstName
Hansen	Ola
Svendson	Tove
Pettersen	Kari

SQL - Select

Persons

P_Id	LastName	FirstName	Address	City
1	Hansen	Ola	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Kari	Storgt 20	Stavanger

SELECT * FROM Persons

Persons

P_Id	LastName	FirstName	Address	City
1	Hansen	Ola	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Kari	Storgt 20	Stavanger

SQL – Select Distinct

SELECT DISTINCT column_name(s)
FROM table_name

Persons

P_Id	LastName	FirstName	Address	City
1	Hansen	Ola	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Kari	Storgt 20	Stavanger

SELECT DISTINCT City FROM Persons

City
Sandnes
Stavanger

SQL Where

SELECT column_name(s)
FROM table_name
WHERE column_name operator value

Persons

P_Id	LastName	FirstName	Address	City
1	Hansen	Ola	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Kari	Storgt 20	Stavanger

SELECT * FROM Persons
WHERE City='Sandnes'

P_Id	LastName	FirstName	Address	City
1	Hansen	Ola	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes

SQL Where

This is correct:

```
SELECT * FROM Persons WHERE FirstName='Tove'
```

This is wrong:

```
SELECT * FROM Persons WHERE FirstName=Tove
```

This is correct:

```
SELECT * FROM Persons WHERE Year=1965
```

This is wrong:

```
SELECT * FROM Persons WHERE Year='1965'
```

Operators

Operator	Description
=	Equal
<>	Not equal
>	Greater than
<	Less than
>=	Greater than or equal
<=	Less than or equal
BETWEEN	Between an inclusive range
LIKE	Search for a pattern
IN	To specify multiple possible values for a column

The AND & OR Operators

Persons

P_Id	LastName	FirstName	Address	City
1	Hansen	Ola	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Kari	Storgt 20	Stavanger

SELECT * FROM Persons WHERE FirstName='Tove' AND LastName='Svendson'

P_Id	LastName	FirstName	Address	City
2	Svendson	Tove	Borgvn 23	Sandnes

The AND & OR Operators

Persons

P_Id	LastName	FirstName	Address	City
1	Hansen	Ola	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Kari	Storgt 20	Stavanger

SELECT * FROM Persons WHERE FirstName='Tove' OR FirstName='Ola'

P_Id	LastName	FirstName	Address	City
1	Hansen	Ola	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes

Combining AND & OR

Persons

P_Id	LastName	FirstName	Address	City
1	Hansen	Ola	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Kari	Storgt 20	Stavanger

SELECT * FROM Persons WHERE LastName='Svendson‘ AND (FirstName='Tove' OR FirstName='Ola')

P_Id	LastName	FirstName	Address	City
2	Svendson	Tove	Borgvn 23	Sandnes

Order by

SELECT column_name(s) FROM table_name
ORDER BY column_name(s) ASC|DESC

Persons

P_Id	LastName	FirstName	Address	City
1	Hansen	Ola	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Kari	Storgt 20	Stavanger
4	Nilsen	Tom	Vingvn 23	Stavanger

SELECT * FROM Persons ORDER BY LastName

P_Id	LastName	FirstName	Address	City
1	Hansen	Ola	Timoteivn 10	Sandnes
4	Nilsen	Tom	Vingvn 23	Stavanger
3	Pettersen	Kari	Storgt 20	Stavanger
2	Svendson	Tove	Borgvn 23	Sandnes

ORDER BY DESC

Persons

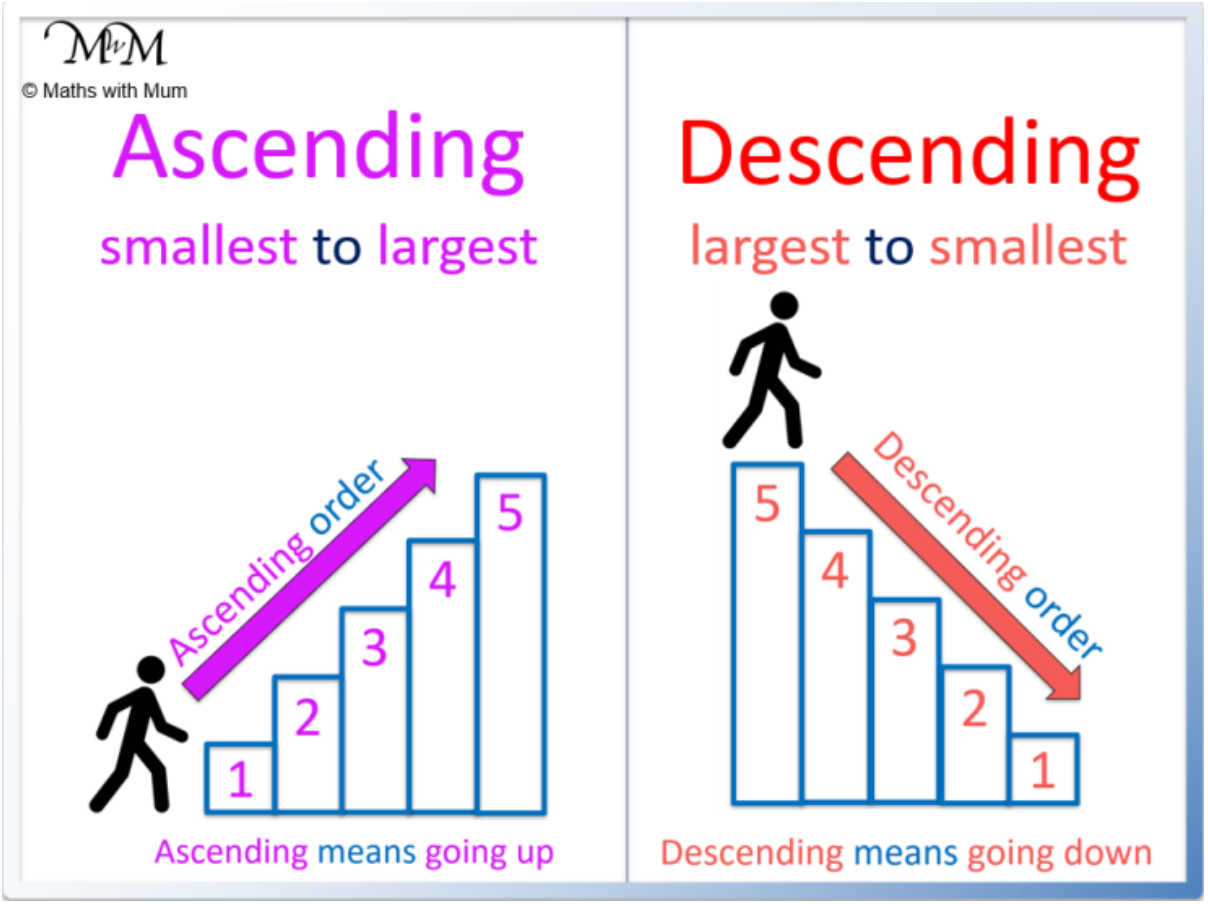
P_Id	LastName	FirstName	Address	City
1	Hansen	Ola	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Kari	Storgt 20	Stavanger
4	Nilsen	Tom	Vingvn 23	Stavanger

SELECT * FROM Persons ORDER BY LastName DESC

P_Id	LastName	FirstName	Address	City
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Kari	Storgt 20	Stavanger
4	Nilsen	Tom	Vingvn 23	Stavanger
1	Hansen	Ola	Timoteivn 10	Sandnes

DESC = Descending

ASC = Ascending



The INSERT INTO Statement

```
INSERT INTO table_name
VALUES (value1, value2, value3,...)
```

```
INSERT INTO table_name (column1, column2, column3,...)
VALUES (value1, value2, value3,...)
```

The INSERT INTO Statement

Persons

P_Id	LastName	FirstName	Address	City
1	Hansen	Ola	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Kari	Storgt 20	Stavanger

INSERT INTO Persons VALUES (4,'Nilsen', 'Johan', 'Bakken 2', 'Stavanger')

P_Id	LastName	FirstName	Address	City
1	Hansen	Ola	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Kari	Storgt 20	Stavanger
4	Nilsen	Johan	Bakken 2	Stavanger

Insert Data บน Columns ที่กำหนด

```
INSERT INTO Persons (P_Id, LastName, FirstName)
VALUES (5, 'Tjessem', 'Jakob')
```

Persons

P_Id	LastName	FirstName	Address	City
1	Hansen	Ola	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Kari	Storgt 20	Stavanger
4	Nilsen	Johan	Bakken 2	Stavanger
5	Tjessem	Jakob		

SQL Update

```
UPDATE table_name
SET column1=value, column2=value2,...
WHERE some_column=some_value
```

SQL Update

Persons

P_Id	LastName	FirstName	Address	City
1	Hansen	Ola	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Kari	Storgt 20	Stavanger
4	Nilsen	Johan	Bakken 2	Stavanger
5	Tjessem	Jakob		

```
UPDATE Persons SET Address='Nissestien 67', City='Sandnes'
WHERE LastName='Tjessem' AND FirstName='Jakob'
```

P_Id	LastName	FirstName	Address	City
1	Hansen	Ola	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Kari	Storgt 20	Stavanger
4	Nilsen	Johan	Bakken 2	Stavanger
5	Tjessem	Jakob	Nissestien 67	Sandnes

SQL Delete

```
DELETE FROM table_name
WHERE some_column=some_value
```

SQL Delete

Persons

P_Id	LastName	FirstName	Address	City
1	Hansen	Ola	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Kari	Storgt 20	Stavanger
4	Nilsen	Johan	Bakken 2	Stavanger
5	Tjessem	Jakob	Nissestien 67	Sandnes

```
DELETE FROM Persons WHERE LastName='Tjessem' AND
FirstName='Jakob'
```

P_Id	LastName	FirstName	Address	City
1	Hansen	Ola	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Kari	Storgt 20	Stavanger
4	Nilsen	Johan	Bakken 2	Stavanger

Delete ทุกแถว

DELETE FROM table_name

or

DELETE * FROM table_name

SQL Like

- SELECT column_name(s)

FROM table_name

WHERE column_name LIKE pattern

- Pattern ในที่นี่คือรูปแบบ เช่น 's%' – ขึ้นต้นด้วย s และตามด้วยอะไรอีกก็ได้
- '%tav%' – ประกอบด้วย tav ในนั้น
- '%end' – จบคำด้วย end ในนั้น
- อ่านต่อได้ใน Wildcards:

https://www.w3schools.com/sql/sql_wildcards.asp

SQL Like

Persons

P_Id	LastName	FirstName	Address	City
1	Hansen	Ola	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Kari	Storgt 20	Stavanger

SELECT * FROM Persons WHERE City LIKE 's%'

P_Id	LastName	FirstName	Address	City
1	Hansen	Ola	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Kari	Storgt 20	Stavanger

SQL Like

Persons

P_Id	LastName	FirstName	Address	City
1	Hansen	Ola	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Kari	Storgt 20	Stavanger

SELECT * FROM Persons WHERE City LIKE '%s'

P_Id	LastName	FirstName	Address	City
1	Hansen	Ola	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes

SQL Like

Persons

P_Id	LastName	FirstName	Address	City
1	Hansen	Ola	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Kari	Storgt 20	Stavanger

- SELECT * FROM Persons WHERE City LIKE '%tav%'

P_Id	LastName	FirstName	Address	City
3	Pettersen	Kari	Storgt 20	Stavanger

SQL In

- SELECT column_name(s) FROM table_name WHERE column_name IN (value1,value2,...)
- สามารถระบุค่าได้หลายค่าในประโยค Where

SQL In

Select * from Persons where Lastname = 'Hansen' or Lastname = Pettersen or Lastname='svendson' or lastname = 'Obama'

Persons

P_Id	LastName	FirstName	Address	City
1	Hansen	Ola	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Kari	Storgt 20	Stavanger

SELECT * FROM Persons WHERE LastName IN ('Hansen','Pettersen','Obama')

P_Id	LastName	FirstName	Address	City
1	Hansen	Ola	Timoteivn 10	Sandnes
3	Pettersen	Kari	Storgt 20	Stavanger

SQL BETWEEN

- SELECT column_name(s) FROM table_name WHERE column_name
BETWEEN value1 AND value2
- เลือกข้อมูลที่อยู่ระหว่างช่วง value1 – value2

SQL BETWEEN

Products

ProductID	ProductName	SupplierID	CategoryID	Unit	Price
1	Chais	1	1	10 boxes x 20 bags	18
2	Chang	1	1	24 - 12 oz bottles	19
3	Aniseed Syrup	1	2	12 - 550 ml bottles	10
4	Chef Anton's Cajun Seasoning	2	2	48 - 6 oz jars	22
5	Chef Anton's Gumbo Mix	2	2	36 boxes	21.35
6	Grandma's Boysenberry Spread	3	2	12 - 8 oz jars	25
7	Uncle Bob's Organic Dried Pears	3	7	12 - 1 lb pkgs.	30
8	Northwoods Cranberry Sauce	3	2	12 - 12 oz jars	40

```
SELECT * FROM Products
WHERE Price BETWEEN 18 AND 20;
```

ProductID	ProductName	SupplierID	CategoryID	Unit	Price
1	Chais	1	1	10 boxes x 20 bags	18
2	Chang	1	1	24 - 12 oz bottles	19

SQL NOT BETWEEN

Products

ProductID	ProductName	SupplierID	CategoryID	Unit	Price
1	Chais	1	1	10 boxes x 20 bags	18
2	Chang	1	1	24 - 12 oz bottles	19
3	Aniseed Syrup	1	2	12 - 550 ml bottles	10
4	Chef Anton's Cajun Seasoning	2	2	48 - 6 oz jars	22
5	Chef Anton's Gumbo Mix	2	2	36 boxes	21.35
6	Grandma's Boysenberry Spread	3	2	12 - 8 oz jars	25
7	Uncle Bob's Organic Dried Pears	3	7	12 - 1 lb pkgs.	30
8	Northwoods Cranberry Sauce	3	2	12 - 12 oz jars	40

```
SELECT * FROM Products
WHERE Price NOT BETWEEN 15 AND 40;
```

ProductID	ProductName	SupplierID	CategoryID	Unit	Price
3	Aniseed Syrup	1	2	12 - 550 ml bottles	10

SQL Alias

- สร้างชื่อใหม่ให้กับ Table หรือ Column
- SELECT column_name FROM table_name AS alias_name
- SELECT column_name1 AS alias_name1, column_name2 AS alias_name2
FROM table_name
- SELECT column_name1, column_name2 + ', ' + column_name3 + ' ' + column_name4 + ', ' + column_name5 AS alias_name
FROM table_name